04-07 第1篇

阅读理解（说明文）

① Have you ever wondered why the sky is blue? Many people think it's because the sky reflects (反射) the color of the ocean. However, this is not true. The real reason has to do with light from the sun and our Earth's atmosphere (大气层).

② Light from the sun looks white to us, but it is actually made up of all the colors of the rainbow: red, orange, yellow, green, blue, indigo (靛青), and violet (紫罗兰色). You can see these colors when sunlight passes through a glass prism (棱镜) or when a rainbow appears in the sky after the rain. Light travels in waves. Different colors of light have different wave lengths. Red light waves are long, while blue and violet light waves are short.

③ Around the Earth is the atmosphere. It is full of gases and dust. When sunlight reaches the Earth's atmosphere, it hits these tiny particles (微粒) in the air. Because blue light travels in shorter, smaller waves, it gets scattered (散射) in all directions by the gases in the air much more than the other colors.

④ This scattering of blue light makes the sky look blue to our eyes. Why not violet, since violet waves are even shorter than blue ones? First, the sun puts out much more blue light than violet light. Second, our human eyes are more sensitive (敏感的) to blue light than to violet light. So we see a blue sky instead of a purple one!

⑤ Next time you look up at a clear blue sky, you can proudly explain the science behind it to your friends.

注释： reflect 反射；atmosphere 大气层；particle 微粒；scatter 散射；sensitive 敏感的

选择题

1. According to the passage, why do some people incorrectly think the sky is blue? A. Because of the white light from the sun. B. Because the sky reflects the color of the ocean. C. Because there are blue gases in the air. D. Because the atmosphere is full of dust.

2. Which color light travels in the longest waves according to paragraph ②? A. Red. B. Blue. C. Green. D. Violet.

3. What happens when sunlight hits the particles in the atmosphere? A. All colors are scattered equally (平等地). B. Only red light is scattered. C. Blue light is scattered more than other colors. D. The particles turn into small rainbows.

4. Why doesn't the sky look violet to us? A. Because there are no violet waves in sunlight. B. Because violet is not a color of the rainbow. C. Because our eyes are more sensitive to blue light and the sun puts out less violet light. D. Because violet waves are too long to reach the Earth.

5. What would be the best title for this passage? A. Why the Ocean is Blue B. How to Make a Glass Prism C. The Colors of the Rainbow D. Why the Sky is Blue

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详细解析

1. 【答案】B

【题型】 细节理解

【定位】 第①段第2-3句 "Many people think it's because the sky reflects (反射) the color of the ocean. However, this is not true."

【解析】 原文提到，很多人错误地认为天空是蓝色的，是因为天空反射了海洋的颜色（reflects the color of the ocean）。因此，这是人们的错误认知。其余选项均不是导致误解的原因。故选B。

2. 【答案】A

【题型】 细节理解判断

【定位】 第②段最后一句 "Red light waves are long, while blue and violet light waves are short."

【解析】 由文中所述，“红光光波长（Red light waves are long），而蓝光和紫光光波短”，故在这几个给出的选项中，红光的光波是最长的。故选A。

3. 【答案】C

【题型】 细节理解判断

【定位】 第③段第3-4句 "When sunlight reaches the Earth's atmosphere... Because blue light travels in shorter, smaller waves, it gets scattered in all directions by the gases in the air much more than the other colors."

【解析】 当阳光遇到大气中的微粒时，文中所描述的具体事实是：之所以天空更显著表现出蓝色，并非平均散射（A），也非只有红光被散射（B），更不是微粒变成彩虹（D），而是蓝光比其他颜色被“散射得更多（scattered in all directions... much more...）”。故选C。

4. 【答案】C

【题型】 细节理解

【定位】 第④段第2-4句 "Why not violet, since violet waves are even shorter than blue ones? First... Second, our human eyes are more sensitive (敏感的) to blue light than to violet light."

【解析】 第四段专门解释了为什么我们看到的天空不是紫色的：一是太阳发出的紫光少于蓝光；二是我们人类的眼睛对蓝光比对紫光更敏感。A、B、D项与原文或者常识相去甚远。故选C。

5. 【答案】D

【题型】 主旨大意题

【定位】 贯穿全文的设问及解答思路。

【解析】 本文从开头的提问“你有没有想过天空为什么是蓝色的？”（Have you ever wondered why the sky is blue?）引入，后面通过分析太阳光成分（第②段）、大气微粒散射蓝光（第③段）、以及人眼敏感度和紫光区别（第④段）揭晓科学原理。因此，围绕说明“天空为何是蓝色”的D项是最佳标题。A、B、C项都是以局部替代整体或风马牛不相及的主题。故选D。